Philipps-Universität Marburg
Fachbereich Mathematik & Informatik

Georg Wille
Volkmar Welker

Probabilistic Combinatorics SS24

Sheet 4

Due: 13. May at 19:00

(i) There is a chess tournament with 40 players with 80 games played, where every two players face each other at most once. Proof:

(i) There are at least 8 players, no two of which played each other. Tipp: Choose cleverly and use (ii).

(ii) Proof: For real numbers $a_1, ..., a_r > -1$ holds:

$$\frac{1}{a_1+1} + ... + \frac{1}{a_r+1} \geq \frac{r^2}{(a_1+1)+...+(a_r+1)}.$$

(2) A subset $\Sigma \subseteq [n]$ is called additively free, if for every two subsets $A, B \subseteq \Sigma$:

$$\sum_{i \in A} i = \sum_{i \in B} i \quad \Rightarrow \quad A = B.$$

(i) Find an additively free subset $\Sigma \subseteq [n]$, such that $|\Sigma| = \lfloor \log_2(n) \rfloor + 1$.

(ii) Use the Chebyshev's inequality to show: There is an upper bound for the cardinality of an additively free subset $\Sigma \subseteq [n]$ given by

$$|\Sigma| \leq \log_2(n) + \log_2 \log_2(n) + 1.$$

(3) Consider the Erdős–Rényi model $G(n, p_n)$ for random generating graphs. Let X_1 be the number of isolated vertices in $G(n, p_n)$. Proof: If $p_n = \frac{c_n \log(n)}{n}$ for some $c_n \gg 1$, then

$$\mathbb{P}(X_1 > 0) = o(1).$$

(4) In the Erdős–Rényi model $G(n, p_n)$ let $p_n = 1/n$. Show: If n is big enough, there is a constant $c > 0$, such that the expected number of vertices of degree ≥ 3 (i.e. the vertex lies in ≥ 3 edges), is at least cn.